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Thought for 5m 10s >

Here is the complete English translation of the PDF's substantive content, with the formulas kept unchanged.  Erdos#1183

## Set Theory Problem Solving

I am currently studying:

Let  $f(n)$  be maximal such that in any 2-colouring of the subsets of  $\{1, \dots, n\}$  there is always a monochromatic family of at least  $f(n)$  sets which is closed under taking unions and intersections. Estimate  $f(n)$ .

Let  $F(n)$  be defined similarly, except that we only require the family be closed under taking unions. Estimate  $F(n)$ . In particular, is it true that

$$F(n) \geq n^{\omega(n)}$$

for some  $\omega(n) \rightarrow \infty$  as  $n \rightarrow \infty$ , and

$$F(n) < (1 + o(1))^n?$$

Don't search the internet; please solve it.

### Thought for 64m 30s

I will first split the problem into two parts: one is the size of a sublattice closed under both unions and intersections, and the other is the case where only union-closure is required. I will first look for general lower bounds, then construct

colourings to obtain upper bounds, and see whether the order of magnitude can be pinned down more clearly.

I will first focus on  $F(n)$ , rewriting "union-closed" in the language of posets / join-semilattices in order to capture its structure; at the same time I will try to obtain an upper bound more concrete than "subexponential," together with a definite lower bound.

A new upper-bound direction has already emerged: if one colours sets according to the parity of their size, then any monochromatic union-closed family has at most  $2^{\lfloor n/2 \rfloor}$  members, so your guess

$$F(n) < (1 + o(1))^n$$

initially looked quite likely to be false. Another line of small-scale brute force computation gives

$$f(1) = F(1) = 1, \quad f(2) = F(2) = 2, \quad f(3) = F(3) = 2, \quad f(4) = F(4) = 3,$$

so I was also testing whether they might in fact equal

$$\left\lceil \frac{n+1}{2} \right\rceil.$$

The small-scale computation was pushed further to  $n = 5$ : we already see

$$f(5) = 3 \quad \text{while} \quad F(5) = 4,$$

so the two quantities start to diverge from here. Another structural line of analysis also pushed

$$f(n)$$

down to

$$f(n) \leq Cn \log n.$$

Write  $\log = \log_2$ . I can prove the following estimates:

$$\frac{n+1}{2} \leq f(n) \leq (4 + o(1)) n \log n,$$

and

$$\frac{n+1}{2} \leq F(n) \leq 2^{(1+o(1))(\log n)^2} = n^{(1+o(1)) \log n}.$$

Therefore, for the half you asked about specifically,

$$F(n) < (1 + o(1))^n,$$

the answer is **yes**.

As for the other half,

$$F(n) \geq n^{\omega(n)},$$

the argument below does not yield it; at present I can only give the linear lower bound  $\frac{n+1}{2}$ .

## 1. First, two obvious lower bounds

Take a maximal chain in the Boolean lattice  $2^{[n]}$ :

$$\emptyset \subset \{1\} \subset \{1, 2\} \subset \cdots \subset [n].$$

This chain has  $n + 1$  sets. Under any 2-colouring, one of the colours appears on this chain at least  $\lceil \frac{n+1}{2} \rceil$  times.

Moreover, any subfamily of a chain is automatically closed under both unions and intersections (because the union is the larger of the two sets, and the intersection is the smaller one).

So

$$f(n) \geq \frac{n+1}{2}.$$

Likewise, a chain is of course also closed under unions, so

$$F(n) \geq \frac{n+1}{2}.$$

## 2. Upper bound for $f(n)$ : $f(n) \leq (4 + o(1))n \log n$

### 2.1 Structure of sublattices

If  $L \subseteq 2^{[n]}$  is closed under unions and intersections, then it is a finite distributive lattice. Standard Birkhoff representation tells us:

- there exists a finite poset  $P$  such that

$$L \cong J(P),$$

where  $J(P)$  is the lattice of ideals of  $P$ ;

- since  $L$  is concretely embedded in the Boolean lattice, the coordinates  $1, \dots, n$  split into pairwise disjoint blocks

$$B_0, B_\infty, \{B_p : p \in P\},$$

such that every set in  $L$  can be written as

$$B_0 \cup \bigcup_{p \in I} B_p, \quad I \in J(P).$$

Therefore, counting sublattices of a Boolean lattice is equivalent to:

first choosing a poset  $P$ , and then assigning the  $n$  coordinates to these blocks.

## 2.2 Width is controlled by size

Let  $|L| = s$ , i.e.  $|J(P)| = s$ . If  $P$  has a  $w$ -element antichain  $A$ , then for every subset  $E \subseteq A$  one gets a different ideal; hence

$$s \geq 2^w.$$

So

$$w(P) \leq \log s.$$

That is, for a sublattice of size  $s$ , the width of the corresponding poset is at most  $\log s$ .

## 2.3 Counting posets of width $\leq w$

Let  $|P| = m$ , with width  $\leq w$ . By Dilworth,  $P$  can be partitioned into  $w$  chains.

First estimate the number of choices for an “ordered chain decomposition”: at most

$$m! w^m.$$

(Give a total ordering, then label each element by a chain; the order within each chain is then determined.)

Now fix two chains

$$C = (c_1 < \dots < c_a), \quad D = (d_1 < \dots < d_b).$$

For each  $c_t$ , the set  $\{j : d_j < c_t\}$  must be an initial segment of  $D$ , while  $\{j : c_t < d_j\}$  must be a final segment; hence the relation between the two chains is determined by two weakly increasing sequences, and the number of possibilities is at most

$$\binom{a+b}{a}^2 \leq 2^{2(a+b)}.$$

Multiplying over all pairs of chains, and noting

$$\sum_{i < j} (a_i + a_j) = (w-1)m,$$

we obtain: after fixing the chain decomposition, the total number of possibilities for all cross-chain relations is at most

$$2^{2(w-1)m}.$$

Thus, the number of posets of width  $\leq w$  on  $m$  labelled elements is at most

$$m! w^m 2^{2(w-1)m}.$$

## 2.4 Then put the coordinates back in

Given such a  $P$ , each coordinate  $x \in [n]$  only needs to decide which block it belongs to:

- $B_0$ ,
- $B_\infty$ ,
- or some  $B_p$  ( $p \in P$ ).

So there are at most

$$(m+2)^n$$

ways.

Therefore, the total number  $N_s$  of sublattices of size exactly  $s$  satisfies (writing  $w = \lfloor \log s \rfloor$ ):

$$N_s \leq \sum_{m \leq n} (m+2)^n m! w^m 2^{2(w-1)m}.$$

Using  $m \leq n$ ,  $m! \leq n^n$ ,  $w \leq \log s$ , and  $2^{2w} \leq s^2$ , we get

$$N_s \leq n(n+2)^n n^n (\log s)^n s^{2n}.$$

That is,

$$\log_2 N_s \leq 2n \log s + 2n \log n + n \log \log s + O(n).$$

## 2.5 Random 2-colouring

Now colour  $2^{[n]}$  randomly with two colours. The probability that a fixed  $s$ -element sublattice is monochromatic is

$$2^{1-s}.$$

So the expected number of monochromatic  $s$ -element sublattices is at most

$$2^{1-s} N_s.$$

The counting above shows that when

$$s \geq (4 + \varepsilon)n \log n,$$

the exponent  $-s$  dominates the remaining terms, so

$$2^{1-s} N_s = o(1).$$

Summing over all  $s \geq (4 + \varepsilon)n \log n$ , this is still  $o(1)$ . Hence there exists a colouring with no monochromatic sublattice of size exceeding  $(4 + \varepsilon)n \log n$ .

Therefore

$$f(n) \leq (4 + o(1))n \log n.$$

In summary:

$$\frac{n+1}{2} \leq f(n) \leq (4 + o(1))n \log n.$$

### 3. Upper bound for $F(n)$ : $F(n) \leq 2^{(1+o(1))(\log n)^2}$

Here we only require closure under unions, so the structure is much freer; however, one can use a very clean VC/Sauer idea.

#### 3.1 A large union-closed family forces a large free join-semilattice

Let

$$B_r^- := \{\emptyset \neq X \subseteq [r]\}.$$

Equipped with the operation “union,” this is the free join-semilattice on  $r$  generators (with the empty set removed).

#### Key lemma.

If a union-closed family  $U \subseteq 2^{[n]}$  has size

$$|U| > \sum_{i=0}^{r-1} \binom{n}{i},$$

then  $U$  must contain a copy of  $B_r^-$ .

**Proof.** By Sauer-Shelah,  $U$  shatters some  $r$ -element set  $T \subseteq [n]$ . Thus for each  $i \in T$ , we can choose  $A_i \in U$  such that

$$A_i \cap T = \{i\}.$$

For each nonempty  $X \subseteq T$ , define

$$U_X := \bigcup_{i \in X} A_i.$$

Since  $U$  is union-closed,  $U_X \in U$ . Moreover,

$$U_X \cap T = X,$$

so these  $U_X$  are pairwise distinct, and satisfy

$$U_X \cup U_Y = U_{X \cup Y}.$$

This is exactly a  $B_r^-$ . Proof complete.

So, as long as one colour class contains no monochromatic  $B_r^-$ , it automatically follows that

$$F(n) \leq \sum_{i=0}^{r-1} \binom{n}{i}. \quad (1)$$

### 3.2 Use random colouring to eliminate large $B_r^-$

We estimate how many copies of  $B_r^-$  there are in  $2^{[n]}$ .

A  $B_r^-$  is determined by  $r$  generators  $G_1, \dots, G_r$ . For each ground element  $x \in [n]$ , one only needs to record which generators contain it; this gives a subset

$$I_x \subseteq [r].$$

So in total there are at most

$$(2^r)^n = 2^{rn}$$

choices. That is, the number of copies of  $B_r^-$  is at most  $2^{rn}$ .

A fixed  $B_r^-$  has  $2^r - 1$  points, and the probability that it is monochromatic is

$$2 \cdot 2^{-(2^r-1)} = 2^{2-2^r}.$$

Hence under random 2-colouring, the expected number of monochromatic  $B_r^-$  is at most

$$2^{rn+2-2^r}.$$

As long as

$$2^r \gg rn,$$

this expectation tends to 0.

Take

$$r = \lceil \log n + 2 \log \log n \rceil,$$

then

$$2^r \asymp n(\log n)^2,$$

hence

$$2^r \gg rn.$$

Therefore there exists a colouring with no monochromatic  $B_r^-$ .

Substituting into (1):

$$F(n) \leq \sum_{i < r} \binom{n}{i}.$$

Using the standard estimate

$$\sum_{i < r} \binom{n}{i} \leq \left(\frac{en}{r}\right)^r,$$

we obtain

$$F(n) \leq 2^{(1+o(1))(\log n)^2} = n^{(1+o(1)) \log n}.$$

In particular,

$$F(n) < (1 + o(1))^n.$$

## 4. Final summary

So what I obtain is:



$$\frac{n+1}{2} \leq f(n) \leq (4+o(1))n \log n$$

and

$$\frac{n+1}{2} \leq F(n) \leq 2^{(1+o(1))(\log n)^2}.$$

Therefore:

- $f(n)$  is at least linear, and at most  $n \log n$ ;
- $F(n)$  is at least linear, and at most quasi-polynomial;
- for your question

$$F(n) < (1+o(1))^n,$$

the answer is yes;

- for your question

$$F(n) \geq n^{\omega(n)},$$

I still cannot prove it; the present method only yields a linear lower bound.

If you would like a more “research note style” version, the next most natural thing for me to do would be to fully expand the “sublattice = weighted ideal lattice” representation in the  $f(n)$  part into an independent chain of lemmas.

来源

